

Reproducibility Checklist

1. General Paper Structure

- 1.1. Includes a conceptual outline and/or pseudocode description of AI methods introduced (yes/partial/no/NA) [yes](#)
- 1.2. Clearly delineates statements that are opinions, hypothesis, and speculation from objective facts and results (yes/no) [yes](#)
- 1.3. Provides well-marked pedagogical references for less-familiar readers to gain background necessary to replicate the paper (yes/no) [yes](#)

2. Theoretical Contributions

- 2.1. Does this paper make theoretical contributions? (yes/no) [no](#)

If yes, please address the following points:

- 2.2. All assumptions and restrictions are stated clearly and formally (yes/partial/no) [NA](#)
- 2.3. All novel claims are stated formally (e.g., in theorem statements) (yes/partial/no) [NA](#)
- 2.4. Proofs of all novel claims are included (yes/partial/no) [NA](#)
- 2.5. Proof sketches or intuitions are given for complex and/or novel results (yes/partial/no) [NA](#)
- 2.6. Appropriate citations to theoretical tools used are given (yes/partial/no) [NA](#)
- 2.7. All theoretical claims are demonstrated empirically to hold (yes/partial/no/NA) [NA](#)
- 2.8. All experimental code used to eliminate or disprove claims is included (yes/no/NA) [NA](#)

3. Dataset Usage

- 3.1. Does this paper rely on one or more datasets? (yes/no) [yes](#)

If yes, please address the following points:

- 3.2. A motivation is given for why the experiments are conducted on the selected datasets (yes/partial/no/NA) [yes](#)
- 3.3. All novel datasets introduced in this paper are included in a data appendix (yes/partial/no/NA) [NA](#)
- 3.4. All novel datasets introduced in this paper will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no/NA) [NA](#)

- 3.5. All datasets drawn from the existing literature (potentially including authors' own previously published work) are accompanied by appropriate citations (yes/no/NA) [yes](#)

- 3.6. All datasets drawn from the existing literature (potentially including authors' own previously published work) are publicly available (yes/partial/no/NA) [yes](#)

- 3.7. All datasets that are not publicly available are described in detail, with explanation why publicly available alternatives are not scientifically satisfying (yes/partial/no/NA) [NA](#)

4. Computational Experiments

- 4.1. Does this paper include computational experiments? (yes/no) [yes](#)

If yes, please address the following points:

- 4.2. This paper states the number and range of values tried per (hyper-) parameter during development of the paper, along with the criterion used for selecting the final parameter setting (yes/partial/no/NA) [partial](#)
- 4.3. Any code required for pre-processing data is included in the appendix (yes/partial/no) [no \(in Code and Data Supplement\)](#)
- 4.4. All source code required for conducting and analyzing the experiments is included in a code appendix (yes/partial/no) [no \(in Code and Data Supplement\)](#)
- 4.5. All source code required for conducting and analyzing the experiments will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no) [yes](#)
- 4.6. All source code implementing new methods have comments detailing the implementation, with references to the paper where each step comes from (yes/partial/no) [partial](#)
- 4.7. If an algorithm depends on randomness, then the method used for setting seeds is described in a way sufficient to allow replication of results (yes/partial/no/NA) [yes](#)
- 4.8. This paper specifies the computing infrastructure used for running experiments (hardware and software), including GPU/CPU models; amount of memory; operating system; names and versions of relevant software libraries and frameworks (yes/partial/no) [partial](#)
- 4.9. This paper formally describes evaluation metrics used and explains the motivation for choosing these metrics (yes/partial/no) [partial](#)
- 4.10. This paper states the number of algorithm runs used

to compute each reported result (yes/no) **yes**

- 4.11. Analysis of experiments goes beyond single-dimensional summaries of performance (e.g., average; median) to include measures of variation, confidence, or other distributional information (yes/no) **yes**
- 4.12. The significance of any improvement or decrease in performance is judged using appropriate statistical tests (e.g., Wilcoxon signed-rank) (yes/partial/no) **no**
- 4.13. This paper lists all final (hyper-)parameters used for each model/algorithm in the paper's experiments (yes/partial/no/NA) **partial**